D-R-1928
March 15, 1978
SUPERSEDING
MIL-R-43370A
June 30, 1969

FEDERAL SPECIFICATION

RAMP, LOADING VEHICLE (DOCKBOARD), COUNTERBALANCED, MECHANICAL, ADJUSTABLE, 20,000 POUND CAPACITY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers self-contained, mechanically operated, counterbalanced, hinged-lip, adjustable, vehicle loading ramps (dockboards).

1.2 Classification. Ramps shall be of the following types and sizes, as specified (see 6.2).

Type A - Recessed; pit mounted.
Type B - Free-standing.

Size I - 8 feet long by 6 feet wide.
Size II - 10 feet long by 6 feet wide.
Size III - 8 feet long by 7 feet wide.
Size IV - 10 feet long by 7 feet wide.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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U.S. Department of Commerce Standard:

CS202-56 - Industrial Lifts and Hinged Loading Ramps.

(Application for copies should be addressed to the U.S. Department of Commerce, National Bureau of Standards, Standards Development Services Section, Technology Building A168, Washington, DC 20234.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Certification. Prior to offering the first production lot of Loading Vehicle Ramps (hereinafter known as "ramps") to the Government for acceptance, the contractor shall furnish, to the contracting officer or his authorized representative, satisfactory evidence of compliance with the applicable requirements of CS202-56. Satisfactory evidence of said compliance shall be a written declaration of same used in conjunction with the contractor's name and address on labels, invoices, sales and technical literature, etc.,; a test report indicating compliance with CS202-56 from a recognized independent testing laboratory acceptable to the Government; or other similar evidence which is deemed acceptable by the Government.

3.2 Design. Ramps shall be self-contained units of mechanically functioning equipment, designed for permanent installation at loading platforms (docks) in accordance with structural characteristics particular to the type and size specified. Ramps shall be constructed primarily of unitized structural steel. Ramp travel surfaces, including the hinged lip-extension, shall be made of a skid-resistant, raised-pattern, floor-plate steel. The ramp design shall be such that all ramp members may be recessed behind the dock face (for Type A), or the forward most projection of the ramp from (for Type B), when the ramp is in the stored position. In the stored position, ramps shall have no vertically projecting members which might impede cross-traffic flow. Ramps shall be counterbalanced, and each ramp shall have a safety device, as specified herein. Safe and readily accessible means shall be provided for the lubrication of ramp components.

3.2.1 Working range. Ramps shall provide a total vertical (inclined) adjustment of not less than 24 inches, divided equally above and below dock level.

3.2.2 Lateral compensation. Ramps shall provide automatic lateral compensation for differences in the side-to-side elevation (i.e. tilt) of a carrier bed during loading and unloading operations. When tested as specified in 4.3.3, such compensation shall accommodate up to a maximum of four inches difference in the side-to-side, vertical elevation of a canted carrier bed, anywhere within the ramp's working range.

3.2.3 Features.

3.2.3.1 Skirts. Ramps shall be provided with skirts (toe guards) on each side made of nominal No. 14 gauge (.0687 inches thick, min.) sheet steel. The skirts shall not interfere with ramp movement, and shall be of sufficient depth to provide full protection throughout the ramps entire operating range (see 6.3).

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3.2.3.2 Hinged-lip. Ramps shall be provided with a retractable hinged-lip extending the entire width of the ramp travel surface. When activated into its use position, the lip shall project not less than twelve inches beyond the dock-bumper faces, at dock level. The lip shall be beveled on all exterior edges.

3.2.3.3 Bumpers. Ramps shall be equipped with two, shock-absorbing bumpers, for appropriate mounting on each side of the ramp travel surface, either on the dock face (for Type A) or on the forward most projection of the ramp frame (for Type B). The bumpers shall be at least four inches thick and shall provide sufficient surface area to adequately protect ramp and dock face from damage resulting from vehicular traffic. Bumpers shall be either the laminated rubber or tire-fabric type.

3.3 Rated capacity. Ramps shall have individual rated capacities of 20,000 pounds. The rated capacity is defined as 100 percent of the "total gross load," 85 percent of which is to be applied to one axle and 15 percent to the second axle of a loading vehicle having a nominal fifty inch wheelbase and a width of 34 to 42 inches, measured from centerline to centerline of tread. The total gross load is defined as a moving load, moving at a velocity of not less than five miles per hour (7.33 ft/sec) onto a ramp positioned at not less than a five percent grade. When ramps are tested as specified in 4.3.1, there shall be no evidence of permanent damage or distortion which prevents or inhibits normal ramp operation.

3.4 Counterbalance. When tested as specified in 4.3.3, the ramp counterbalance feature shall provide automatic, vertical compensation to changes in the carrier bed elevation created by carrier spring expansion and compression during loading and unloading operations. During the applicable portion(s) of that test, firm and continuous contact shall be maintained between the hinged lip and the carrier bed.

3.5 Safety device. An automatic safety device shall be provided to stop, within the limits cited below, a downward movement of the ramp should a loss of contact occur between the hinged lip and the carrier bed as a result of accidental removal of the carrier bed. When working three or more inches above dock level, the ramp shall be stopped at, or within one inch below, dock level. When working at less than three inches above dock level and below, the ramp downward movement shall be stopped within four inches. The safety device shall be effective with any load up to a maximum of 20,000 pounds, anywhere within the working range of the ramp. Design of the safety device shall be such that following an occurance of its use, any manual adjustment necessary for resumption of normal operation shall be accomplishable from above dock level. When the ramp is tested as specified in 4.3.2, ramp movement shall be stopped within the limits specified above, and normal operation shall be resumed as specified above. As a result of testing specified in 4.3.2, no damage shall occur which would prevent or inhibit normal operation of the ramp.

3.6 Finish. Prior to painting, exterior surfaces of the ramp shall be thoroughly cleaned and dried; exterior surfaces shall be free from mill scale, oil, grease, dirt, and rust. The exterior surfaces shall be painted as soon as practicable after cleaning. They shall be painted with suitable primer(s) and baked enamel(s) in accordance with the manufacturer's standards practice. The color shall be the manufacturer's standard color. The finished surfaces shall be smooth, evenly coated, and free from runs, sags, blisters, flaked areas, imbedded material, rust and other defects. Surfaces of components and assemblies not normally painted shall be cleaned and prepared according to the manufacturer's standard practice, to assure proper performance of intended function.

3.7 Connections, welds, and castings. Holes for rivets and bolts shall be accurately punched, drilled, reamed, or cast, as applicable; adjacent edges shall be free from burrs, fins, and other projections which might inhibit the proper seating of the connecting device(s). Welded joints shall form a complete and continuous fusion of the welded materials; they shall be smooth, and free from pits, holes cracks, and other fissures. Rivet heads shall be full, neatly made, concentric with rivet holes, and in full contact with the surface of the member. Excessive upsetting of rivets to fill holes shall not be acceptable. Castings shall be free of blowholes, porosity, cracks, foreign material, and other injurious defects (see 4.2.1).

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3.8 Operation. The ramp operational cycle shall include the following elements:

(a) actuation from the stored position by a manual control, integral to the ramp, that releases the locking mechanism and allows the travel surface to rise to its maximum height;
(b) extension and locking into travel position of the hinged lip at, or during the rise to, the maximum height, not inhibited by the presence of a carrier backed up against the dock bumpers;
(c) descent of the travel surface, and engagement of the hinged lip with the carrier bed, under influence of an individual weighing not more than 150 pounds, walking the length of the travel surface;
(d) employment of the lateral compensation and counter balance mechanisms in accordance with 3.2.2 and 3.4, respectively;
(e) use of the ramp, starting from the stored position, to facilitate handling of an end load positioned below dock level (this element shall be demonstrable without the use of special tools or need for the operator to reach beneath the ramp.);
(f) retraction of the hinged lip from the carrier bed at the operators' discretion, regardless of the carrier position; automatic retraction of the lip to its out-of-use position upon removal of the carrier bed support;
(g) automatic locking (into stored position) of the ramp travel surface when it is manually removed from a use position and returned to dock level.

When tested as specified in 4.3.3, said (operational cycle) elements shall be satisfactorily demonstrated without evidence of malfunction and without the use of any tool or special device* that is not an integral part of the ramp. Such demonstration need not proceed in the specific sequence noted above.

*NOTE: "Device", as used here, does not refer to any "testing equipment" - e.g., loading vehicle, carrier, freight containers, and other items - which might be basic to the performance of said test.

3.9 Parts and technical manual. One technical manual, including a comprehensive parts list, shall be enclosed in a waterproof envelope and securely attached to each ramp prior to shipment. All information in the manual not specifically related to the equipment to which it is attached shall be so noted. The manual shall provide the necessary instructions for installation and proper use and maintenance of the ramp.

3.10 Nameplate. Each ramp shall have a corrosion-resistant metal plate securely attached, in a readily visible location on some exterior surface of the ramp. The following information shall be indented or embossed on the plate;

(a) description of the equipment;
(b) name of the manufacturer;
(c) serial and model number;
(d) rated capacity;
(e) shipping weight;
(f) date of manufacture (month/year).

3.11 Installation. As part of the delivered ramp, the contractor shall provide all materials (including mechanical and connecting devices, structural forms, diagrams and written instructions, etc.) necessary for the proper installation and initial maintenance of the ramp(s) at the Government's site of intended use. Unless otherwise specified (see 6.2), the contractor shall be responsible for, and bear all costs attendant to, such installation and initial maintenance. When specified (see 6.2), said installation procedures and costs shall include those necessary for pit construction or alteration at the Government's site. When installation, etc. by the contractor is required, it shall be supervised by the contractor's authorized representative and shall be subject to approval by the Government. The initial maintenance shall include the necessary adjustments, lubrications, and other steps as needed to render the ramp(s) operational.

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3.12 Workmanship. Workmanship shall be first class throughout. All exposed ramp surfaces shall be free of slivers, burrs, sharp edges, and other injurious defects. Ramps shall be free of defects which affect ramp durability, detract from ramp appearance or which may impair ramp usability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling. Unless otherwise specified in this section, sampling for examinations and tests shall be conducted in accordance with MIL-STD-105.

4.2.1 Visual and dimensional examination. Ramps shall be visually examined for the defects listed in Table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects, end items

Defect	Major	Minor
Ramp is not mechanical, counter balanced type	X	
Type or size (see 1.2) not as specified	X	
Travel surface not raised-pattern, plate steel	X	
Sheet steel skirts (toe guards) missing or not as specified	X	
Toe guard does not provide protection through-out entire operating range	X	
Toe guard or other ramp member protrudes above surface of ramp when in the stored position	X	
Hinged lip does not project at least 12 inches beyond dock-bumper faces, when fully extended	X	
Exterior edges of hinged lip not beveled	X	
Ramp members cannot be recessed level with or behind the dock face (for Type A) or forward most projection of the frame (for Type B)		X
Shock absorbing bumpers missing or not as specified	X	
Exterior surfaces not painted	X	
Finish is unevenly applied, blistered, peeled, chipped, cracked; contains imbedded foreign material, or incorporates other defects related to appearance		X
Any evidence of rust or corrosion	X	
Machined parts not free of burrs, fins, or other sharp projections	X	
Welded and cast parts not free of pits, holes, cracks, or other lesions	X	
Fastening devices not secured		X
Rivet heads not full, neatly made and in full contact with surface		X
Technical manual including parts list, missing or not as specified		X
Nameplate missing or not as specified		X

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4.2.2 Sampling for tests.

4.2.2.1 Repetitive loading test. A single sample shall be tested as specified in 4.3.1. The sample shall be a properly installed and fully operational ramp, selected at random from the first production lot offered to the Government for acceptance. Failure of the sample to comply with the test requirements shall be cause for rejection of the lot. Compliance with the test specified in 4.3.1 need only be ascertained once during the term of any one contract, unless essential design, mechanical, or material features (of the ramp) are altered during the contract life. If such alterations are instituted, the contractor shall notify the Government, and sampling and testing as before shall be conducted, at the Government's discretion.

4.2.2.2 Quality assurance tests. Ramps offered to the Government for acceptance shall be tested as specified in 4.3.2 and 4.3.3. The inspection level shall be S-3 with an AQL of 2.5 expressed in terms of percent defective.

4.3 Tests.

4.3.1 Repeated loading test. A ramp selected as specified in 4.2.2.1 shall be actuated and positioned appropriately on the bed of a suitable carrier. The travel surface shall be at an elevation of not less than 5 percent. The ramp shall then be subjected to 150 roll-over cycles (see 6.3), with 33,000 pounds (total gross load), under the conditions of load distribution, loading-vehicle specification, and velocity specified in 3.3. Following completion of the test in the manner specified, the ramp shall be examined for compliance with the requirements of 3.3.

4.3.2 Safety device test. The ramp safety device shall be tested twice, at two ramp elevations, each time supporting a weight of not less than 20,000 pounds. At each elevation, the hinged lip shall be fully engaged with the bed of a suitable carrier, with the weight positioned at the approximate center of the travel surface. The point(s) of reference for measurement purposes shall be the loading dock surface, at each leading edge on either side of the ramp (for type B ramps, corresponding points shall be chosen on the ramp frame). The first ramp elevation shall be three or more inches above the loading dock surface; the second elevation shall be at some intermediate point between dock level and 12 inches below dock level. In each case, the carrier bed shall be moved abruptly out from under the ramp surface. Precise measurements shall be made before and after each test to determine compliance with the requirements of 3.5. Compliance with the "operational capability" requirements of 3.5 shall be satisfied by successful completion of the test specified in 4.3.3.

4.3.3 Operational test. Ramps tested as specified in 4.3.2 shall be made to demonstrate all functional elements designated in 3.8 (a) through (g), in the manner specified therein. All elements shall be demonstrated with a suitable carrier backed up against the dock bumpers. For the purpose of examining the lateral compensation feature (3.2.2 and 3.8(d)), the carrier shall be canted to an out-of-level position with respect to the horizontal surface of the ramp (with a loading vehicle inside the carrier) in such a way that one corner of the ramp lip remains in contact with the carrier bed. The loading vehicle shall then be driven from inside the carrier out onto the dock. To demonstrate the counter balance feature (3.4 and 3.8(d)), sufficient numbers of appropriately sized and weighted articles shall be moved to and from the carrier and the dock to produce the needed expansion and compression of the carrier springs. Any malfunction which prevents the satisfactory demonstration of an operational element, or which inhibits, the travel to and from carrier and dock by the loading vehicle shall constitute failure of this test.

4.4 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be identified in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipments fully prepared for shipment. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components, such as loose strapping.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Each complete ramp shall be preserved (when applicable), packaged, and packed for domestic or export shipment in accordance with normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling, and storage, to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and to be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The loading vehicle ramps (dockboards) covered by this specification are used to span and compensate for space and height differentials between loading docks and freight carriers, to facilitate safe, efficient, freight transfer. The type A ramp is recessed into a preformed pit in the loading dock; the type B ramp normally requires no dock preparation and is permanently affixed to the face or curb angle of the dock.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) Responsibility for installation and initial maintenance (see 3.11).
(d) When pit construction or alteration is required (see 3.11).
(e) Marking required (see 5.2).

6.3 Definitions. The following terms, as used in this specification, are defined as indicated.

carrier: A wheeled, enclosed trailer or container that, when attached to a heavy-duty truck or van, is used to carry bulk freight over long distances.

operating range: the maximum extent of vertical space, above and below dock level, through which the ramp's inclined travel surface can rise and descend; compare to "working" range.

roll-over cycle: the movement of a load from the loading dock, onto the ramp, into the carrier, and back out onto the dock.

stored position: the normal position of a ramp when not in use; locked in place at dock level, with the hinged lip not extended; "cross-traffic" position and "out-of-use" position are synonomous terms.

unitized structural steel: structural steel forms (e.g. beams, plates, rods, channels, etc.) designed to carry out a specific function when assembled.

working range: the extent of vertical space, above and below dock level, for which a ramp is designed to provide compensation during loading and unloading operations.

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6.4 Supersession. Federal specification D-R-1928 supersedes military specification MIL-R-43370A, Ramp, Loading Vehicle, Adjustable, Mechanically Operated, 20,000 Pound Capacity, June 30, 1969.

Military Custodians:

Army - ME
Navy - SA
Air Force - 84

Military Coordinating Activity:

Army - ME

Review Activity:

Army - AT, SM,
DGSC - GS

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

USPA-POS
VA-VOC
HEW-FEC
NASA-JFK
GSA-FSS,PCD

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